

Outline

- 1) Hypothesis testing
- 2) Neyman - Pearson Lemma
- 3) Uniformly most powerful tests

Hypothesis Testing

Model $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$

Null hypothesis $H_0 : \theta \in \Theta_0$ ("default")

Alternative hyp. $H_1 : \theta \in \Theta_1$

Hypotheses should be disjoint: $\Theta_0 \cap \Theta_1 = \emptyset$

and exhaustive $\Theta_0 \cup \Theta_1 = \Theta$

Want to use data X to learn which includes θ

Inductive behavior:

We either reject H_0 (conclude $\theta \in \Theta_1$), or
fail to reject H_0 (no conclusion)

("Accept H_0 " OK as technical term, but may confuse)

$H_{0,1}$ called simple if $\Theta_{0,1} = \{\theta_{0,1}\}$ composite o.w.

Ex $X \sim N(\theta, 1)$

$H_0 : \theta \leq 0$ vs $H_1 : \theta > 0$

(composite vs. composite)

$H_0 : \theta = 0$ vs $H_1 : \theta \neq 0$

(simple vs composite)

Ex $X_1, \dots, X_n \sim P$ $Y_1, \dots, Y_m \sim Q$

$H_0 : P = Q$ vs $H_1 : P \neq Q$

(composite vs. composite)

Critical Function

Can describe a test formally by its
critical function (a.k.a. test function)

$$\phi(x) = \begin{cases} 0 & \text{accept } H_0 \\ \pi \in (0,1) & \text{reject w.p. } \pi \\ 1 & \text{reject } H_0 \end{cases}$$

In practice, randomization rarely used ($\phi(x) = \{0,1\}$)
(In theory, simplifies discussions.)

A non-randomized test partitions $\mathcal{X}$ into

$$R = \{x : \phi(x) = 1\} \quad \text{rejection region}$$

$$A = \{x : \phi(x) = 0\} \quad \text{acceptance region}$$

Usually defined via test statistic $T(x) \in \mathbb{R}$

We say ϕ rejects for large $T(x)$ if

$$\phi(x) = \begin{cases} 0 & T(x) < c \\ 1 & T(x) > c \\ \gamma \in (0,1) & T(x) = c \end{cases} \quad (\text{if randomized})$$

for critical threshold $c \in \mathbb{R}$

T chosen to discriminate well between H_0, H_1

Significance Level and Power

Two types of errors

- 1) Type I error: H_0 true but we reject
- 2) Type II error: H_0 false but we don't rej.

Usual goal is to minimize P_{H_1} (Type II error),
while controlling P_{H_0} (Type I error) $\leq$ fixed α

Note if $H_{0,1}$ composite, " $P_{H_{0,1}}$ " is not a well-defined prob.

Power function:
$$\beta_\phi(\theta) = E_\theta[\phi(x)]$$
$$= P_\theta[\text{Reject } H_0]$$

fully summarizes test's behavior

Goal: multiple objectives multiple constraints
maximize β_ϕ for $\theta \in \mathcal{H}_1$, subject to $\beta_\phi \leq \alpha$ for $\theta \in \mathcal{H}_0$.

ϕ is a level- α test ($\alpha \in [0,1]$) if $\sup_{\theta \in \mathcal{H}_0} \beta_\phi(\theta) \leq \alpha$

Ubiquitous choice is $\alpha = 0.05$

["Most influential offhand remark in history of science"]

Question: Can we find ϕ^* that maximizes power everywhere on the alternative at once?

Z test

Ex Test statistic $Z(X) \sim N(0, 1)$ (very common)

Def Upper α quantile $z_\alpha = \Phi^{-1}(1-\alpha)$, $\Phi = N(0, 1)$ cdf

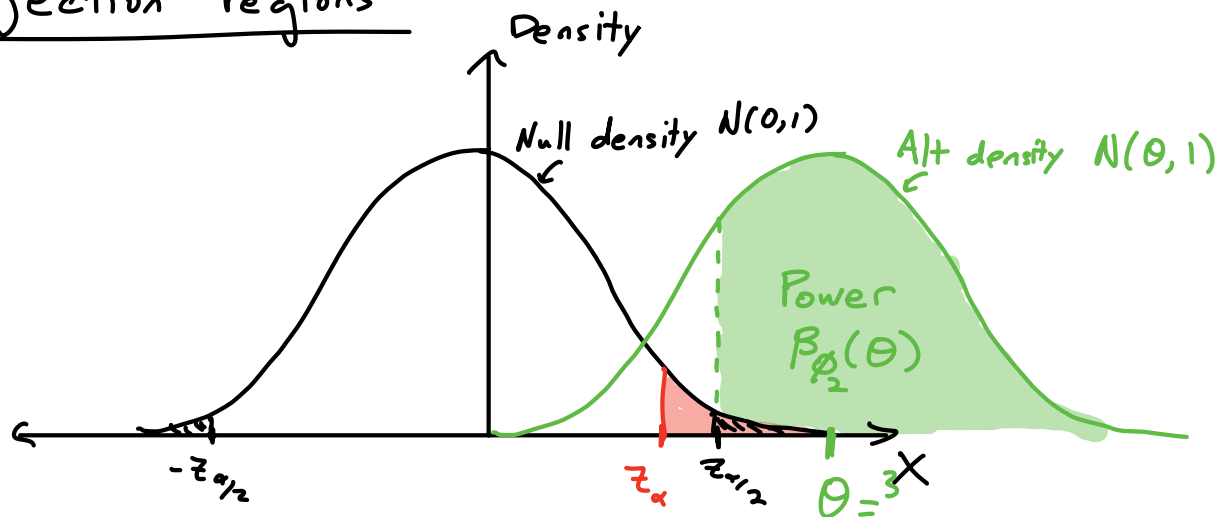
1-sided Z-test: $H_0: \theta \leq 0$ vs $H_1: \theta > 0$

$$\phi_1(x) = 1\{x > z_\alpha\}$$

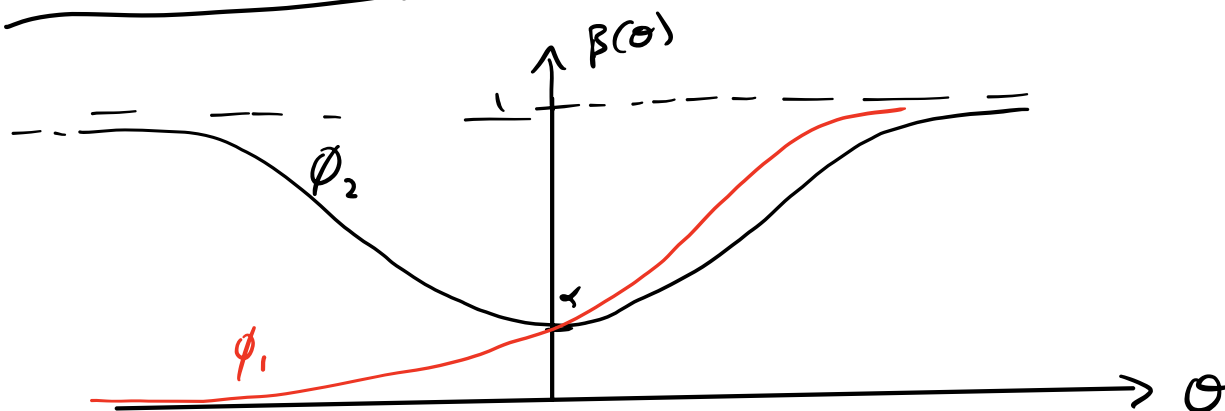
2-sided Z-test: $H_0: \theta = 0$ vs $H_1: \theta \neq 0$

$$\phi_2(x) = 1\{|x| > z_{\alpha/2}\} \quad (\text{Could also use } \phi_1(x))$$

Rejection regions



Power functions:



Likelihood Ratio Test

Simple vs simple: $H_0: X \sim P_0$ vs $H_1: X \sim P_1$

Densities p_0, p_1 wrt dominating measure μ (e.g. $P_0 + P_1$)

Optimal test rejects for large values of

Likelihood ratio: $LR(x) = p_1(x)/p_0(x)$

Likelihood ratio test (LRT):

$$\phi^*(x) = \begin{cases} 1 & LR(x) > c \\ \gamma & LR(x) = c \\ 0 & LR(x) < c \end{cases}$$

c, γ chosen to make $\mathbb{E}_0 \phi^*(x) = \alpha$

Intuition:

Power under H_1 : $\max \int_{\mathcal{R}} p_1(x) d\mu(x)$

"Bang"

Sig. budget: $\int_{\mathcal{R}} p_0(x) d\mu(x) \leq \alpha$

"Buck"

Spend fixed α budget on x values
that deliver greatest bang/buck

Neyman-Pearson

Theorem (Neyman-Pearson Lemma)

LRT with significance level α is optimal for testing $H_0: X \sim p_0$ vs. $H_1: X \sim p_1$

Proof We are interested in maximization problem

$$\begin{array}{ll} \text{maximize} & \mathbb{E}_1[\phi(x)] \\ \phi: X \rightarrow [0,1] & \text{s.t. } \mathbb{E}_0[\phi(x)] \leq \alpha \end{array}$$

Lagrange form:

$$\begin{aligned} \text{maximize} \quad & \mathbb{E}_1[\phi(x)] - \lambda \mathbb{E}_0[\phi(x)] \\ &= \int \phi(x) (p_1(x) - \lambda p_0(x)) d\mu(x) \\ &= \int \phi(x) \left(\frac{p_1(x)}{p_0(x)} - \lambda \right) dP_0(x) \end{aligned}$$

$$\text{Solution(s):} \quad \phi(x) = \begin{cases} 1 & \text{if } LR > \lambda \\ 0 & \text{if } LR < \lambda \\ \text{arbitrary} & \text{if } LR = \lambda \end{cases}$$

$\Rightarrow \phi^*$ maximizes Lagrangian for $\lambda = c$

Consider any other test $\tilde{\phi}(x)$, $\mathbb{E}_0 \tilde{\phi}(x) \leq \alpha$

$$\begin{aligned} \mathbb{E}_1 \tilde{\phi} &\leq \mathbb{E}_1 \tilde{\phi} - c \mathbb{E}_0 \tilde{\phi} + c\alpha \\ &\leq \mathbb{E}_1 \phi^* + c \mathbb{E}_0 \phi^* + c\alpha \\ &\leq \mathbb{E}_1 \phi^* \end{aligned}$$

$$c(\alpha - \mathbb{E}_0 \tilde{\phi}) \geq 0$$

ϕ^* maxes Lagrangian

$$c(\alpha - \mathbb{E}_0 \phi^*) \geq 0$$

Ex $X \sim \text{Binom}(n, \theta)$

Test $H_0: \theta = .5$ vs $H_1: \theta = .51$ at level $\alpha = 0.05$

$$p_\theta(x) = \binom{n}{x} \theta^x (1-\theta)^{n-x}$$

$$\Rightarrow \frac{p_{.51}(x)}{p_{.5}(x)} = \frac{.51^x (.49)^{n-x}}{.5^n} \propto_x \left(\frac{.51}{.49} \right)^x \nearrow \text{in } x$$

Reject for large $LR(X) \Leftrightarrow$ Rej. for large X

$\leadsto$ test stat X ,

threshold $c_\alpha = 95^{\text{th}}$ %ile of $\text{Binom}(n, .5)$

X discrete: $P_{H_0}(X > c_\alpha) < \alpha$ (since $2^{-n} \nmid .05$)

Randomize to "top off" error budget:

$$\text{Set } \gamma = \frac{\alpha - P(X > c_\alpha)}{P_{H_0}(X = c_\alpha)} \leadsto P_{H_0}(\text{Reject } H_0) = \alpha$$

In practice just reject for $X > c_\alpha$

conservative test (sig. level $< \alpha$)

What about testing $H_0: \theta = 0.5$ vs $H_1: \theta = 0.508$?

Same test! $\left(\frac{.508}{.492} \right)^x$ also $\nearrow$ in X

Uniformly most powerful tests

General H_0, H_1 (simple or composite)

Def If $\phi^*(x)$ is level- α and for any other level- α test ϕ we have

$$\mathbb{E}_\theta \phi^* \geq \mathbb{E}_\theta \phi \quad \forall \theta \in (1),$$

then ϕ^* is uniformly most powerful (UMP)

Def Assume $\mathcal{P} = \{P_\theta : \theta \in \Theta \subseteq \mathbb{R}\}$ has densities p_θ .

$\mathcal{P}$ has monotone likelihood ratios (MLR) in $T(X)$

if $\frac{p_{\theta_2}(x)}{p_{\theta_1}(x)}$ is non-decreasing function of $T(x)$,

whenever $\theta_1 < \theta_2$ ($\frac{c}{0} := \infty$ if $c > 0$, $\frac{0}{0}$ undefined)

Ex. 1-param exponential family

$$X_1, \dots, X_n \stackrel{iid}{\sim} p_{\eta}(x) = e^{\eta^T(x) - A(\eta)} h(x)$$

$$\frac{p_{\gamma_1}(x)}{p_{\gamma_0}(x)} = \exp \left\{ (\gamma_1 - \gamma_0) \sum_i T(x_i) - n(A(\gamma_1) - A(\gamma_0)) \right\}$$

$$\pi \text{ in } T(X) = \sum_i T(x_i)$$

$\Rightarrow$ LRT rejects for large T if $z_1 > z_0$

11 small T if $\gamma_1 < \gamma_0$

Theorem Assume $\mathcal{P}$ has MLR in $T(X)$, and consider testing $H_0: \theta \leq \theta_0$ vs $H_1: \theta > \theta_0$, for $\theta_0 \in \Theta \subseteq \mathbb{R}$

If $\phi^*(x)$ rejects for large $T(x)$,

ϕ^* is UMP at level $\alpha = \mathbb{E}_{\theta_0} \phi^*(X)$

Proof

Consider any other level- α test ϕ , any $\theta_1 > \theta_0$

ϕ is level- α for $H_0: \theta = \theta_0$ vs $H_1: \theta = \theta_1$

$\frac{p_{\theta_1}(x)}{p_{\theta_0}(x)}$ non-decr in $T(x)$ by assumption

$\Rightarrow \phi^*$ is LRT, $\beta_{\phi^*}(\theta_1) \geq \beta_{\phi}(\theta_1)$

Note $\beta_{\phi^*}(\theta_1) \geq \alpha$ for $\theta_1 > \theta_0$ (compare to $\phi(X) \equiv \alpha$)

Remains to show $\beta_{\phi^*}(\theta) \leq \alpha$ for $\theta < \theta_0$

Consider testing $H_0: \theta = \theta_0$ vs $H_1: \theta = \theta_1$ for $\theta_1 < \theta_0$

Then $\bar{\phi}(X) = 1 - \phi^*(X)$ (reject for small T)

is LRT at level $\mathbb{E}_{\theta_0} \bar{\phi}(X) = 1 - \alpha$

$\Rightarrow 1 - \alpha \leq \beta_{\bar{\phi}}(\theta_1) = 1 - \beta_{\phi^*}(\theta_1)$ for $\theta_1 < \theta_0$ $\square$

Remark We also showed ϕ^* minimizes $\mathbb{P}_{\theta}(\text{Type I error})$ for $\theta < \theta_0$ (among tests with $\mathbb{E}_{\theta_0} \phi(X) = \alpha$)